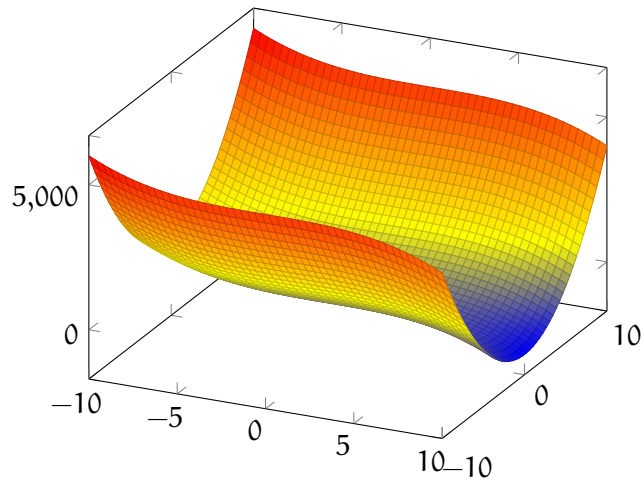


Consider the function

$$f(x_1, x_2) = 50x_1^2 - x_2^3$$

illustrated in the figure below.



1. Show that the point

$$x^* = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

satisfies the necessary optimality conditions.

2. Show that this point is neither a local minimum nor a local maximum. *Hint: Consider the directions*

$$d_1 = \begin{pmatrix} 0 \\ 1 \end{pmatrix} \text{ and } d_2 = \begin{pmatrix} 0 \\ -1 \end{pmatrix}.$$